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Espasandín, L., M. Coll, and V. Sbragaglia. 2024. Distributional range shift of a marine fish relates to a geographical gradient of emotions among recreational fishers. Ecology and Society 29(1):21. https://doi.org/10.5751/ES-14793-290121

Research

Distributional range shift of a marine fish relates to a geographical gradient of emotions among recreational fishers

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ABSTRACT. As the effects of climate change increase, distributional range shifts of species are also expected to be magnified, necessitating a better understanding of their social-ecological implications for the adaptive management of fisheries and biodiversity conservation. In this paper, we focused on the human dimensions of recreational fisheries in the context of an ongoing distributional range shift of a target species. Specifically, we mined data on YouTube from recreational anglers and spearfishers targeting the white grouper (*Epinephelus aeneus*), a species expanding northwards in the northwestern Mediterranean Sea (Italy, France, and Spain). We retrieved 453 videos from Italy and Spain. We analyzed the social engagement of the videos (i.e., number of views, likes, and comments) and applied sentiment analysis to all the comments posted on these videos. Results showed that social engagement is overall higher for spearfishers than anglers. We documented an overall positive polarity and positive emotions in the comments of the posted videos, but specific negative polarity and negative emotions were more common in angling videos than in spearfishing ones. Most importantly, we detected a significant positive correlation between the emotions of joy and surprise and the latitude at which white grouper was caught. This result suggests that recreational fishers may respond to the arrival of the white grouper by showing more joy and surprise at higher latitudes where the species is rare than at lower latitudes where the species is common. Our study illustrates how digital data from social media can be used to monitor social-ecological interactions, such as tracking species distributional range shifts and the human responses to them, with potential management implications. Specifically, these results may be informative to adapt necessary tailored-management actions by improving engagement with fishers and enhancing more effective communication strategies, finally evoking environmental stewardship.

Key Words: *climate change; digital data; emotions; sentiment analysis, social media; species distributional range shift*

INTRODUCTION

Climate change is one of the main drivers of species distributional range shifts (Parmesan and Yohe 2003, Perry et al. 2005, Bellard et al. 2012, Poloczanska et al. 2013, García Molinos et al. 2016, Pinsky et al. 2020). Species shift their latitudinal range to respond to changes in the environment, such as sea warming, and to stay within their preferred environmental conditions (Francour et al. 1994, Cheung et al. 2009, 2013, Burrows et al. 2011, Fogarty et al. 2017). Moreover, within the expected increase of current rates of climate change (Peters et al. 2013, IPCC 2021), distributional range shifts of marine species are also expected to be magnified in the upcoming decades (Cheung et al. 2009, García Molinos et al. 2016). Species distributional range shifts not only affect marine biodiversity and ecosystem function (e.g., trophic interactions; Cheung et al. 2009, Philippart et al. 2011, Pinsky et al. 2020), but they also have social and economic implications (e.g., on food security and human well-being; Allison et al. 2009, Pecl et al. 2017, Bonebrake et al. 2018, Ojea et al. 2020). In fact, marine ecosystems can be studied as complex social-ecological systems where the social and ecological subsystems are highly interconnected and interactive (Ostrom 2009). There is a critical importance of quantitatively measuring interconnections and feedbacks between the social and ecological subsystems in near-real-time (e.g., changes in fish distribution and the response of fishers), and incorporating this understanding into adaptive management in response to climate change (Perry et al. 2010, Ommer et al. 2012, Melbourne-Thomas et al. 2023).

Important and complex feedbacks exist between the marine ecological system and the fishing system (Berkes et al. 1998, Berkes 2011, Perry et al. 2010, Ojea et al. 2020), such as climate-driven marine species distributional range shifts leading to fishing effort reallocation or changes in catch composition (Pinsky and Fogarty 2012, Lam et al. 2016). As an example, Icelandic fishers adapted to the northward expansion of the northeast Atlantic mackerel (*Scomber scombrus*) by reallocating their fishing effort and beginning fishing this new species (Spijkers and Boonstra 2017). The situation led to disputes around the fishing quotas between the European Union (EU), Norway, Iceland, and the Faroe Islands (Spijkers and Boonstra 2017), originating what has been referred to as the "Mackerel Wars." Monitoring these social-ecological feedbacks in fisheries, such as the human dimensions of fishers (e.g., behavioral or emotional responses) toward changes in fish distribution, may give managers early warning of an environmental issue (e.g., the arrival or disappearance of a species) so they can begin planning relevant conservation measures. The ongoing digital revolution may provide a unique opportunity to quantitatively monitor in near-real-time social-ecological feedbacks such as societal attention reactions (Jarić et al. 2023) or emotional responses to environmental issues such as climate change. In this context, recreational fishing stands out as a prime example, given it is a globally widespread activity highly exposed to the impacts of climate change (Townhill et al. 2019), coupled with the pervasive integration of digital platforms within its culture (Lennox et al. 2022).

Human dimensions research in recreational fisheries aims to understand fishers' thoughts and actions (e.g., values, perceptions, or behavioral responses), as well as the interconnections and feedbacks between the social and ecological subsystems (Arlinghaus et al. 2013, 2017, Hunt et al. 2013, Ward et al. 2016). Some of the human dimensions of recreational fisheries to adapt to climate change and species distributional

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range shifts can be effectively understood by the increasing use of social media and digital platforms (Pecl et al. 2019, Townhill et al. 2019), which are embedded within recreational fishers' culture (Sbragaglia et al. 2020, Vitale et al. 2021). Recent research has suggested that the activity of recreational fishers on social media can contribute to understanding climate change effects on the ecological (Sbragaglia et al. 2021a) and social (Sbragaglia et al. 2022) subsystems separately. Moreover, social media may also be useful for providing integrated information on the social-ecological system as a whole. For example, Allison et al. (2023) explored whether user-generated content on Facebook has the potential to encourage pro-environmental behavior among recreational anglers and highlighted how fisheries managers and scientists can facilitate this process (see also Sbragaglia et al. 2023a for another example). Consequently, analysis on social media could help managers gather quantitative social-ecological indicators on how the arrival of new species, or invasive alien species caused by climate change, is perceived by recreational fishers and plan necessary tailored-management actions. For example, sentiment analysis applied to textual content posted by recreational fishers on social media can be used to monitor the changes in their emotions toward the arrival of new species, and consequently help managers and policy makers to adapt with new regulations (e.g., fostering sustainable harvesting of the new and positively perceived species or eradication of invasive alien species if they may have negative impacts).

The Mediterranean Sea is considered a miniature ocean (Lejeusne et al. 2010) that is warming faster than other areas of the world (Lionello and Scarascia 2018, Salat et al. 2019). Moreover, it is an important marine biodiversity hotspot (Myers et al. 2000) and, as such, species distributional range shifts represent a widespread problem for biodiversity conservation and fisheries management (Katsanevakis et al. 2014, Azzurro et al. 2019). We focused this study on the white grouper, *Epinephelus aeneus* (Geoffroy Saint Hilaire, 1817), because it is a target species for both recreational anglers and spearfishers in our study area and there is robust evidence that this species is undergoing a northward distributional range shift (Glamuzina et al. 2000, Dulčić et al. 2006, Riutort 2012, Đođo et al. 2016, Pollard et al. 2018, Bo et al. 2020, Sbragaglia et al. 2023b). Most importantly, previous studies showed that emerging digital methodologies (e.g., YouTube) could be used to track the ongoing distributional range shift of the white grouper in Italy from 2011 to 2017 (Sbragaglia et al. 2021a, 2023b) and characterize social-ecological interactions and feedbacks (i.e., societal interest toward this species distributional range shift; Sbragaglia et al. 2023b). In summary, the ongoing distributional range shift of the white grouper represents a potentially useful case study for testing near-real-time monitoring of an ecological change and its interaction with the human dimensions of recreational fisheries (i.e., emotions). This is key because integrating biological and social near-real-time data is essential for transitioning toward a management approach fluid in space and time, adapted to the resources and users that we seek to manage (Maxwell et al. 2015, Melbourne-Thomas et al. 2022).

The objective of this study is to quantify the emotions of recreational fishers on social media in the context of the ongoing distributional range shift of a target species, the white grouper. Before addressing the primary research question of the study, we first performed a descriptive analysis to assess how human dimensions (i.e., social engagement, frequency of words, and frequency of sentiments) differ between recreational anglers and spearfishers and provide a better understanding of the community of recreational fishers targeting the white grouper. First, we quantified differences in the social engagement of videos posted by recreational anglers and recreational spearfishers. Previous studies showed a species-specific response of recreational fishers in terms of social engagement on YouTube, which could be related to the relative interest in targeting different species (Sbragaglia et al. 2020, 2022). Second, we quantified differences in the frequency of the words and their associated polarity and emotions in the comments posted on videos by recreational anglers and spearfishers. This descriptive analysis provides a better understanding of the social-ecological system of the northwestern Mediterranean recreational fisheries targeting the white grouper. Finally, we asked the primary research question of the study about whether there were differences in emotions and polarity (e.g., joy or surprise) across a latitudinal gradient. Given the white grouper is a target and valued species for recreational fishers and that it is arriving in new areas, and following what Angulo and Courchamp (2009) illustrated about the public valuing rarer species more than common ones, we would expect the same pattern for recreational fishers. This would translate to more positive emotions (e.g., joy or surprise) in response to videos showing the catch of the white grouper at higher latitudes where this species is still rare.

METHODS

Ethical aspects

The data we mined from YouTube are publicly available. However, data privacy concerns and ethical principles associated with human-subject research must be carefully considered when using social media data (Zimmer 2010, Di Minin et al. 2021). We followed recent recommendations for the responsible use of social media data in research (Monkman et al. 2018, Di Minin et al. 2021, Sbragaglia et al. 2021b), considering data privacy concerns and aiming to ensure compliance with the European Union's (EU) General Data Protection Regulation (GDPR). Specifically, we minimized the data by discarding all but the required information and pseudonymized the data by replacing IDs (e.g., channel title, channel ID). We kept all data related to personal information in one dataset, while the rest of the data presented in the paper were stored in a separate dataset. Moreover, all the results are presented in an aggregated format and representative comments were adapted from the original comments (i.e., translated and partially paraphrased) to prevent such information from allowing the identification of the online content used in this study. The methodological approach and data management were approved by the Spanish National Research Council ethical committee (reference number 215/2020).

Data mining

We analyzed videos posted on YouTube by recreational fishers targeting the white grouper from 2010 to 2020 in Italy, France, and Spain. We systematically mined the data using the YouTube Data API (v3), following the steps reported in previous studies (Sbragaglia et al. 2020, 2021a, Correia et al. 2021). We extracted the data from YouTube's API in January 2021 for each country of interest using the common name of the species in the three

different languages (Italian: "cernia bianca"; Spanish: "mero blanco"; French: "mérou blanc"). This approach helped to narrow the results to the study area but also captured homonyms and other non-relevant content (Correia et al. 2017), and thus, data required careful validation (see below). We first compiled a raw dataset with the title and description of videos together with social engagement indicators. These indicators are the total number of views, likes, and comments on the videos posted by both recreational anglers and spearfishers, and they provide three different measures of the social engagement. Then, we manually cross-checked the data to assign the videos to two different categories such as recreational angling and spearfishing, and excluded the videos that were not related to the purpose of the study (e.g., not related to the target species, not showing the catch of the target species, not related to the target countries, or duplicates of previously published videos). We resolved the location of the catch shown in the video first at a general level (e.g., region within a country such as the Balearic Islands or Sardinia), and then when possible at the specific locations together with corresponding geographic coordinates. Data mining was done with the R software (https://www.r-project.org/; version 4.0.3) using "jsonlite" (J. Ooms 2014, *unpublished manuscript*, https://doi.org/10.48550/arXiv.1403.2805), "lubridate" (Grolemund and Wickham 2011), and "curl" (https://cran.r-project.org/web/packages/curl/index.html) packages.

Analysis of social engagement, comments, and sentiment analysis

As mentioned above, we used three different social engagement indicators: the total number of views, likes, and comments. We did that for each one of the groups (i.e., recreational anglers and recreational spearfishers). We also checked if the longevity of the videos on the platform was correlated with the mentioned measures of social engagement.

Regarding the comments, we systematically mined the text of all the public comments associated with the videos previously identified and validated. After downloading the comments with the same approach reported above for the metadata of videos, we transformed the text into tokens (i.e., individual word units), and we removed stop words (meaningless words) as well as some signs without implicit meaning. Then, we analyzed the frequency of occurrence of tokens and their associated polarity and emotions in comments of videos by recreational anglers and spearfishers.

For the sentiment analysis, we analyzed the differences in the polarity and emotions of the comments posted by recreational anglers and spearfishers according to Saif Mohammad's NRC Emotion lexicon (https://saifmohammad.com/WebPages/NRC-Emotion-Lexicon.htm), using the available reference dictionaries in the different languages of the countries with important modifications as follows. The NRC emotion lexicon is a list of words and their associations with two polarities (negative and positive) and eight emotions (anger, fear, anticipation, trust, surprise, sadness, joy, and disgust). The polarity (or valence) refers to the orientation of the expressed sentiment in the text and it can be positive or negative. The output of the sentiment analysis was a dataset with one YouTube comment in each row and 10 columns: two for the polarity and eight for the emotions.

Once the sentiment analysis was completed and in order to address the primary research question, we quantified the differences in emotions and polarity across a latitudinal gradient. To achieve that, we associated the latitude of the catch shown in the video with each emotion or polarity of the comments (i.e., the latitude is a property of the video and not of the comments). We used a modified dictionary already tailored to recreational fisheries (Sbragaglia et al. 2022). In particular, we added some missing words and modified the emotions and polarity associated with the most frequent words used by recreational fishers. The modification of the original dictionary was implemented with a focus group approach with expert recreational fishers. Briefly, we organized a workshop with 10 recreational fishers (Sbragaglia et al. 2022) and co-produced the new dictionary by linking words and emotions with the use of Plutchik's wheel of emotions (Plutchik 1980). We ran all the analyses in R (version 4.0.3) and used the packages "quanteda" (Benoit et al. 2018) and "syuzhet" (Jockers 2020).

Statistical analysis

We used Generalized Linear Models (Nelder and Baker 2006) to estimate differences between recreational angling and recreational spearfishing videos in the social engagement indicators (total number of views, likes, and comments). We estimated differences in the proportion of most used words, polarity, and emotions within comments using a two-tailed z-test. We used the Kendall rank correlation coefficient (Abdi 2006) to evaluate a possible geographical gradient in sentiments (primary research question). To do that, we calculated the mean of each sentiment score (polarity or emotions) of all the comments belonging to videos with the same latitude. The Kendall rank correlation coefficient was also applied to examine the correlation between the variance in the mean sentiment score and the latitude, aiming to establish the consistency of the observed gradient in cases where a significant correlation was identified. In all cases, we used a 95% confidence interval. We ran all the analyses in R (version 4.0.3).

RESULTS

The majority of videos that were identified and validated were from Italy (415; 198 related to angling and 217 to spearfishing), while only 16 videos were from Spain (all of them related to spearfishing). We did not identify useful videos from France (Fig. 1). Given the absence of French videos and the small size of the Spanish sample, we decided to group up the Spanish and Italian videos and not differentiate the results per country. Regarding social engagement (Fig. 2), the number of views related to videos posted by recreational fishers was significantly higher in spearfishing videos than in angling ones (Rate Ratio = 2.67 (2.04–3.48); $\chi^2 = 49.49$; $p < 0.001$; Fig. 2A). The same pattern was found regarding the number of likes (Rate Ratio = 3.18 (2.38–4.24); $\chi^2 = 57.32$; $p < 0.001$; Fig. 2B) and comments (Rate Ratio = 4.29 (3.12–5.88); $\chi^2 = 74.76$; $p < 0.001$; Fig. 2C). Regarding the analysis of correlation between the longevity of the videos and the social engagement (i.e., number of views, likes, and comments), we only found a significantly positive correlation in the videos posted by Italian recreational anglers ($p < 0.05$, tau = 0.14), while the rest were not significantly correlated or even negatively correlated.

Regarding the comments on the videos, we found a total of 3384 comments (552 for angling and 2832 for spearfishing) in a total of 282 videos (100 for angling and 182 for spearfishing). The quantitative analysis of the content of the comments identified a total of 8722 tokens (2354 for angling and 6368 for spearfishing). The frequency analysis of the occurrence of tokens and their associated polarity and emotions revealed several findings. First,

Fig. 1. Map showing the occurrence of *Epinephelus aeneus* in the Mediterranean Sea. Black triangles represent all available occurrences on GBIF excluding social media data (https://doi.org/10.15468/dl.npbj63). Colorful triangles represent those GBIF occurrences retrieved from digital data (see also Fig. S1 in Sbragaglia et al. 2021a). Grey triangles represent four more occurrences from scientific records (Glamuzina et al. 2000, Dulčić et al. 2006, Đođo et al. 2016, Bo et al. 2020). Colorful circles represent the occurrences that were obtained in this study. The colorful representations are shown with different colors according to the year the YouTube video was posted (i.e., as a proxy of the year of catch; see text for more details).

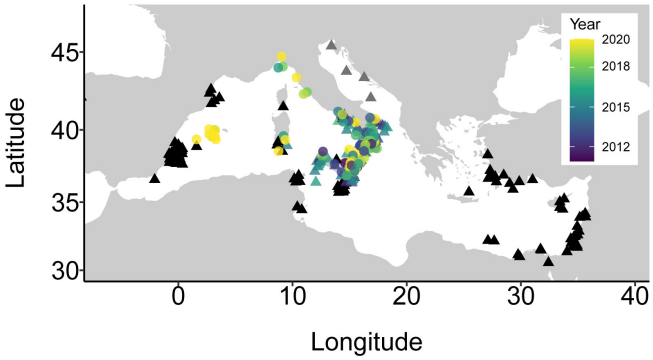


Fig. 2. Barplots of the social engagement, in Italy and Spain, for both angling and spearfishing together with standard error of the mean: (A) number of views; (B) number of likes; (C) number of comments. The total number of videos that were identified and validated was 431 (198 related to angling and 233 to spearfishing). Significant differences are reported according to the results of the GLM models (***: $p < 0.001$).

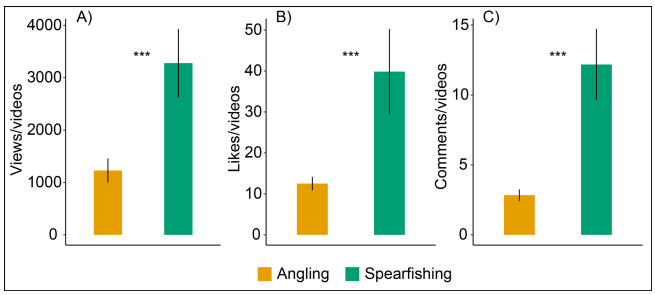
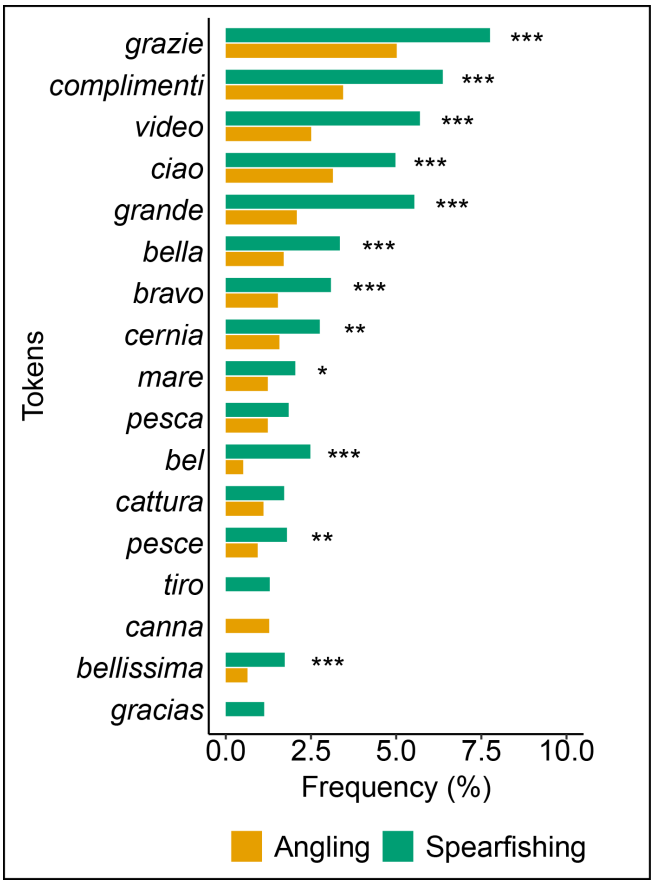


Fig. 3. Frequency of the most common tokens used in the comments on videos posted by recreational anglers and spearfishers. The total of words scored was 8722 tokens (2354 for angling and 6368 for spearfishing). Tokens are reported in Italian and Spanish, and their relative translation is presented between parenthesis as follows: *grazie*/*gracias* (thank you); *complimenti* (congratulations); *video* (video); *grande* (great); *ciao* (hello); *bella* (wonderful); *bravo* (bravo); *cernia* (grouper); *tiro* (shot); *canna* (fishing road); *bel* (wonderful); *mare* (sea); *pesca* (fishing); *cattura* (capture); *pesce* (fish); *bellissima* (wonderful). Significant differences are reported according to the results of the z-test (*: $p < 0.05$; **: $p < 0.01$; ***: $p < 0.001$).



some of the most common tokens in both groups (Fig. 3) were "thank you" ("grazie" in Italian; "gracias" in Spanish), "congratulations" ("complimenti"), "video," "hello" ("ciao"), "great" ("grande"), "beautiful" ("bella" and "bellissima"), or "grouper" ("cernia"). Interestingly, the token "fishing rod" ("canna") appeared only in comments of videos by recreational anglers, and the word "tiro" ("shot") appeared only in comments of videos by recreational spearfishers. In general, these tokens showed a significantly higher ($p < 0.05$) frequency in comments on spearfishing videos than on angling ones (Fig. 3). Second, the sentiment analysis associated with these tokens showed a prevailing positive polarity (Fig. 4). Moreover, there were differences in polarity between comments related to videos posted by recreational anglers and spearfishers (Fig. 4). In particular, we found that positive polarity was significantly lower ($p < 0.001$) in the comments related to angling videos (84.42%) than in spearfishing ones (90.62%). Regarding the emotions, we found that spearfishers indicated a significantly higher ($p < 0.05$) frequency of positive emotions, such as joy (25.60%) and surprise (14.50%), in comparison to videos posted by anglers (21.64% and 12.80%, respectively; Fig. 5). In contrast, comments on videos posted by anglers indicated a significantly higher ($p < 0.05$) frequency of negative emotions, such as fear (4.57%), disgust (2.40%), and anger (3.74%), in comparison to videos posted by spearfishers (2.46%, 1.21%, and 1.36%, respectively; Fig. 5).

Fig. 4. Frequency of polarity of emotions computed with sentiment analysis using a modified version of Saif Mohammad's NRC Emotion lexicon (https://saifmohammad.com/WebPages/NRC-Emotion-Lexicon.htm) for both comments on videos posted by recreational anglers and spearfishers (total polarity scored by anglers = 1117; total polarity scored by spearfishers = 5195. Significant differences are reported according to the results of the z-test (***: $p < 0.001$).

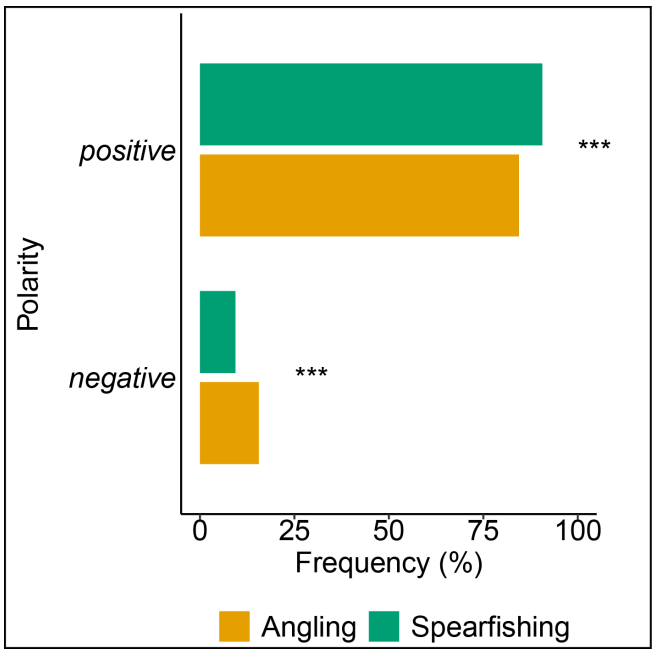
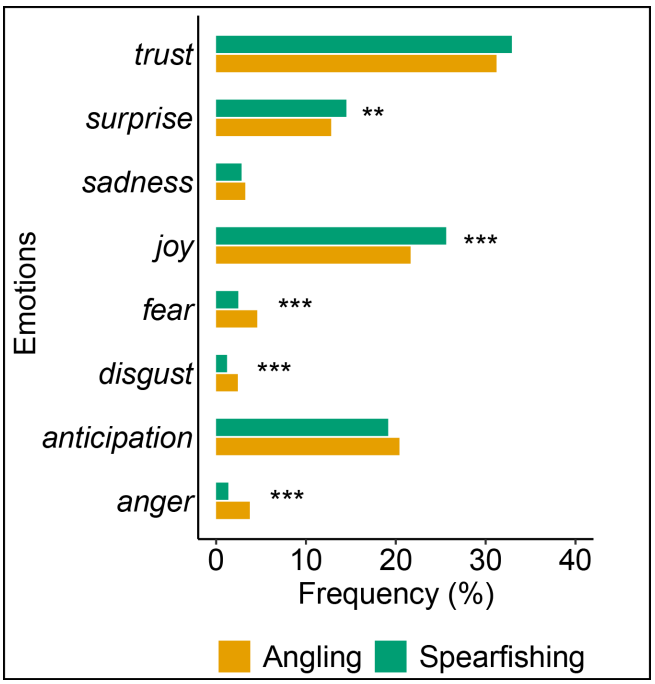


Fig. 5. Frequency of emotions computed with sentiment analysis using a modified version of Saif Mohammad's NRC Emotion lexicon (https://saifmohammad.com/WebPages/NRC-Emotion-Lexicon.htm) for both comments on videos posted by recreational anglers and spearfishers (total sentiments scored by anglers = 2407; total sentiment scored by spearfishers = 11,711). Significant differences are reported according to the results of the z-test (*: $p < 0.05$; ***: $p < 0.001$).



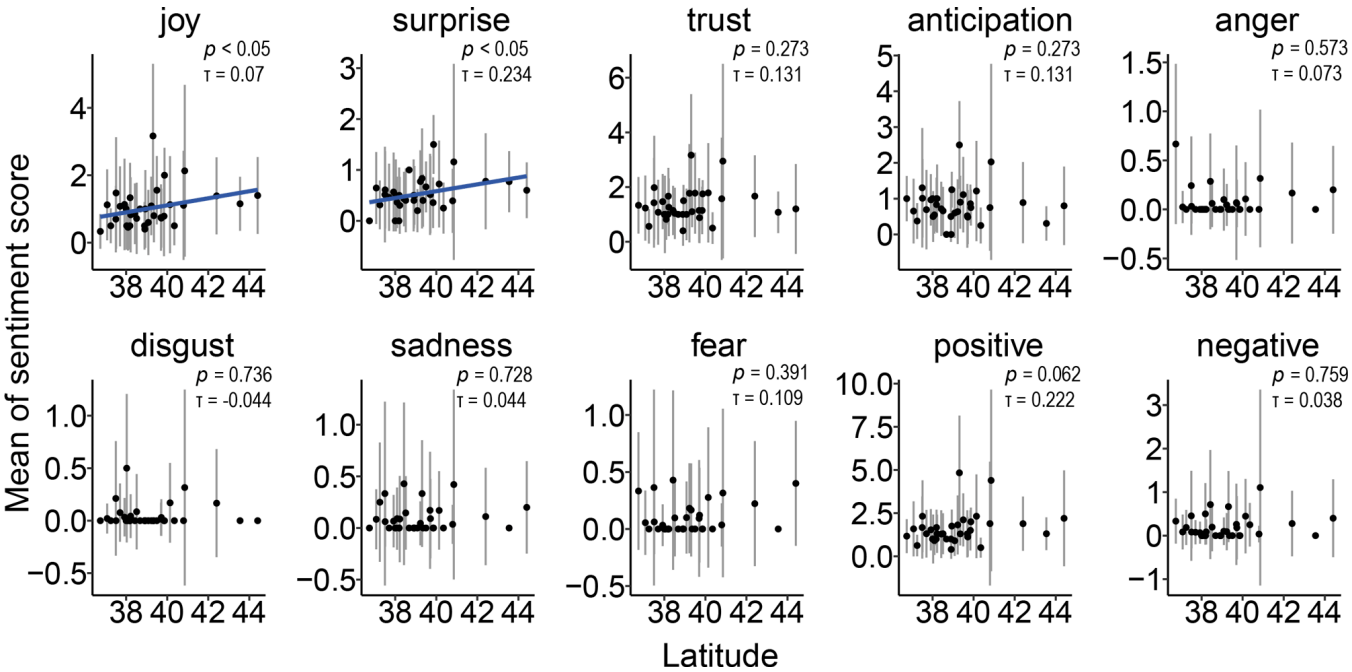
Most importantly, we found that the emotions of joy and surprise showed a significantly positive ($p < 0.05$) correlation with latitude, which means that videos triggered more joy and surprise when showing the capture of a white grouper at higher latitude (Fig. 6; primary research question). In addition, the correlation between latitude and the positive polarity was close to being significant ($p = 0.062$; Fig. 6). Regarding the analysis of the correlation between the variance in the mean sentiment score and the latitude, our findings revealed no significant correlation across all emotions and polarities examined. This absence of correlation is visually depicted in Figure 6, where the variance in the mean sentiment score is illustrated for each latitudinal point, as denoted by the grey error bars.

DISCUSSION

Social media may provide a unique opportunity to quantitatively measure and near-real-time monitor social-ecological feedbacks in the context of recreational fisheries and species distributional range shifts. In the first instance, we found a higher social engagement for videos published by recreational spearfishers than those published by recreational anglers and observed differences in the frequency of most used words in comments. We also found that comments associated with spearfishing videos had a higher frequency of positive polarity and positive emotions (e.g., joy or surprise) and a lower frequency of negative polarity and negative emotions (e.g., fear or sadness) compared with those associated with recreational angling videos. Finally, the core result of our study related to the primary research question showed a significantly positive correlation between both joy and surprise and the latitude, which means that there was an increase of these emotions in the comments related to the videos showing catches of the white grouper at latitudes where the white grouper is expanding its distributional range in recent years.

First, we found that videos posted by recreational spearfishers received more views, likes, and comments than those published by recreational anglers. Although we cannot be sure about the public that engaged with the videos, we assume that most of the social engagement with videos was triggered by either recreational spearfishers or recreational anglers. Previous literature shows a congruent result (i.e., higher social engagement in videos posted by recreational spearfishers) when analyzing videos posted by Italian recreational fishers targeting the common dentex (*Dentex dentex*; Sbragaglia et al. 2020). However, another study focusing on the bluefish (*Pomatomus saltatrix*) showed the opposite (i.e., higher social engagement in videos posted by recreational anglers; Sbragaglia et al. 2022). Therefore, the most plausible interpretation of our result is that the differences between recreational fishing groups are species-specific, which means that recreational spearfishers may be more engaged in fishing the white

Fig. 6. Scatter plot between the latitude and the different emotions and sentiment polarities. We calculated the mean for each one of the sentiment scores of all those comments that presented the same latitude, finding comments for 35 different latitudes (N = 35). Then we applied the Kendall rank correlation coefficient (Abdi 2006) with these mean sentiment scores and the correspondent latitude. The grey error bars represent the variance in the mean sentiment score in each latitude. The blue linear regression line is shown when the correlation is significant. The *p*-value (*p*) and the tau (τ) are reported according to the results of Kendall's rank correlation test.



grouper than anglers. This would explain the absence of videos posted by Spanish recreational anglers targeting the white grouper, which is still a rare species in the area. Additionally, other factors, such as a general preference for YouTube among recreational spearfishers over anglers (Vitale et al. 2021), could contribute to explaining this absence. Altogether, this suggests the necessity for a more refined characterization of the profile of recreational fishers who actively share content on YouTube. Moreover, this result emphasizes the importance of considering species-specific interactions among different recreational fishing groups in the context of climate change and species distributional range shifts.

Second, the analysis of the most frequent words used in the comments suggested that different themes are discussed when the community of recreational fishers interacts and engages with the posted videos. Some of these themes involve expression of congratulations and compliments for the video content, the captured species, and/or the performance of the activity. Some tokens appear to be exclusively used by either recreational anglers or spearfishers as shown in previous studies (Sbragaglia et al. 2022). Most importantly, the outcome of the sentiment analysis showed an overall positive polarity associated with tokens. This result could be related to the fact that the white grouper is a native and valued target species for recreational fishers, so a positive response from them was an expected outcome. It could be also related to the general tendency of individuals to convey positive sentiments on social media (Reinecke and Trepte 2014, Toivonen et al. 2019) and the inherent tendency of such videos to draw in a like-minded community of recreational anglers and spearfishers. Focusing on the differences between the two groups of recreational fishers, the fact that comments on spearfishing videos had a more frequent positive polarity compared with angling ones is worth discussion. This result agrees with a recent study by Sbragaglia et al. (2022), where the higher frequency of negative emotions on angling videos was suggested to be related to the more frequent criticisms. A possible interpretation is that recreational spearfishers are more satisfied than anglers, which is also supported by other studies showing that spearfishers declare more activity satisfaction than anglers (Pita et al. 2018, Gordoa et al. 2019). We did not focus on comparing sentiments among different countries, but there is an important aspect related to the absence of French recreational fishing videos targeting the white grouper. It is possible that French recreational fishers were not comfortable posting videos of the white grouper on social media because the situation of fishing groupers in France was controversial during the study period. The white grouper has been observed in France where fishing "groupers" in general is not allowed. However, the existing legislation during the study period did not specifically forbid the fishing of the white grouper.

The core result of our study, related to the primary research question, showed, to the best of our knowledge, the first quantification of spatial-temporal changes in recreational fishers' emotions and polarity in response to the distributional range shift of a target species. In particular, the most interesting result was that, with the arrival of the white grouper to northern locations, where this species is still rare, recreational fishers express more

positive emotions such as joy and surprise. This result is in line with the conclusions of Angulo and Courchamp (2009) about people valuing rarer species more than common ones, and it meets our predictions because the white grouper is a native and target-valued species for recreational fishers. However, it is important to consider that a new species is perceived according to many factors. One of these factors may be the different ways of perceiving novelty depending on the social group. For example, tourists, managers, and local residents can differ in their perceptions of non-native or invasive species (Kueffer and Kull 2017). One example of mixed perceptions is the rainbow trout (*Oncorhynchus mykiss*), a species introduced in South Africa in 1876 that was positively perceived by some part of the population but negatively perceived by some scientists and conservationists (Shackleton et al. 2019b). Similarly, a species can be positively and negatively perceived within the same group. For example, the bluefish is both negatively and positively perceived by Italian recreational fishers; it is negatively perceived because of its invasiveness, but at the same time it is positively perceived because of its voraciousness and predatory behavior (Sbragaglia et al. 2022), attributes that contribute to the quality of the recreational fishing experience. These examples provide an overview of the complexity of measuring and predicting the changes in human dimensions. Indeed, there is no clear social-ecological framework for the improved characterization and prediction of societal attention reactions and emotional responses to species distributional range shifts. This would be essential because emotions are precursors of intentions and behavior (Manfredo 2008), originating feedback loops in the ecological subsystem and management (Arlinghaus et al. 2013, Hunt et al. 2013, Ward et al. 2016, Wang et al. 2018). The geographical gradient of emotions among recreational fishers in response to the distributional range shift of the white grouper presented in Figure 6 provides managers with a quantitative indicator of perception toward the species. This information may help to adapt necessary tailored-management actions by improving engagement with fishers and enhancing more effective communication strategies, ultimately evoking environmental stewardship (Kelly et al. 2022, Allison et al. 2023, Pecl et al. 2023). In this context, the approach presented here could be used to support management decisions in Italy or Spain by using, for example, the gradient of positive emotions as a proxy of recreational fishing effort on the newly arrived white grouper and plan appropriate conservation actions (i.e., fostering sustainable harvesting through bag limits or seasonal closure of fisheries). Indeed, in November 2023, it was proposed to amend the French legislation regarding white grouper fishing (actual legislation protects groupers, but not the white grouper), adding a ban on newly arrived white grouper for conservation reasons (see the proposed modification of the French prefectural decree no. 2013357-0001).

Some aspects must be considered when interpreting and advancing the research approach presented here, as reported in previous studies (Jarić et al. 2020a, Sbragaglia et al. 2020, 2021b). First of all, further research should consider using this approach with a wider number of species, both range-extended and non-range-extended, and in different areas. This would help to better interpret results related to changes in human dimensions across latitudinal gradients. Moreover, the representativeness of digital data concerning the whole population of recreational fishers is still not characterized. Recent studies suggested that a small proportion of recreational fishers share their data on YouTube, and they tend to be more avid and spend more money than the ones that do not share their catches on social media (Vitale et al. 2021). In the same study, Vitale et al. (2021) found that recreational fishers sharing their content on social media are younger than the ones not sharing it. This may be attributed to the younger age of social media users compared to non-users (Anderson and Jiang 2018) and might be considered as a potential explanatory variable in future studies. For example, recreational fishers who take up fishing after the arrival of a new species might not be as surprised to encounter such species as older fishers. An important limitation in using digital data from social media is that YouTube is a dynamic cultural system where the data can change (e.g., data loss and deletion), limiting the replicability of the study. Furthermore, the geographic location of videos is not always obvious, as happens with the comments where the geographic location is impossible to ascertain. It must also be considered that we selected some keywords for downloading the data from YouTube using its application program interface and this could return only a subsample of all the available information. From a methodological perspective, quantitative text and sentiment analysis present several linguistic challenges (Toivonen et al. 2019). For example, some of the obstacles in analyzing the content of the comments of this study were that some words were used with different meanings and, therefore, may be associated with different emotions (Toivonen et al. 2019; S. M. Mohammad 2020, *unpublished manuscript*, https://doi.org/10.48550/arXiv.2011.03492). We tackled this challenge by creating a customized emotion dictionary for a recreational fishing context (Sbragaglia et al. 2022), and future studies should consider reviewing it according to specific case studies.

CONCLUSION

Our study evaluated a novel quantitative approach that may contribute to the near-real-time monitoring of the human dimension related to a species distributional range shift in the context of recreational fisheries. Social media could open a unique avenue in providing effective quantification of social-ecological feedbacks and thus contribute to informing and guiding managers in the application of effective and adaptive management strategies (Ladle et al. 2016, Becken et al. 2017, Toivonen et al. 2019, Fink et al. 2020). Given the limited accessibility and detectability in aquatic ecosystems (Katsanevakis et al. 2012), this approach stands out as a not excessively time-consuming methodology that can be implemented with limited resources while providing near-real-time data (Jarić et al. 2020b).

In the context of species distributional range shifts, social media may provide managers with valuable information about the perception of a newly arrived species. This information may help to adapt necessary tailored-management actions by improving engagement with users and enhancing effective communication strategies, ultimately leading to environmental stewardship (Allison et al. 2023, Pecl et al. 2023). Although our study exclusively focused on the distributional range shift of a native species in the context of recreational fishing, the research approach presented here could be extended to other contexts, such as monitoring the perception of invasive alien species (Shackleton et al. 2019a, Kapitza et al. 2019, Jarić et al. 2021) or the attitudinal response toward management measures and anticipate reactions (Jarić et al. 2020a).

Author Contributions:

V.S. conceived the study; L.E. collected and analyzed the data; all the authors interpreted the data, and L.E. led the writing of the manuscript with inputs from all authors. All authors gave final approval for publication.

Acknowledgments:

We are extremely thankful to Ricardo A. Correia for developing the R language script for searching videos, and downloading data and comments from the YouTube API. V.S. is supported by a "Ramón y Cajal" research fellowships (RYC2021-033065-I) granted by the Spanish Ministry of Science and Innovation. The authors acknowledge the Spanish government through the "Severo Ochoa Centre of Excellence" accreditation to ICM-CSIC (#CEX2019-000928-S) and partial funding from the European Union's Horizon 2020 research and innovation program under grant agreements No 869300 (FutureMares) and No 101059877 (GES4SEAS).

Data Availability:

The data/code that support the findings of this study are available on request from the corresponding author, L.E. None of the data/code are publicly available because they may contain information that could compromise the privacy of social media users. Ethical approval for this research study was granted by CSIC with 215/2020 as the approval number.

LITERATURE CITED

Abdi, H. 2006. The Kendall rank correlation coefficient. Encyclopedia of Measurement and Statistics 2:508-510.

Allison, E. H., A. L. Perry, M.-C. Badjeck, W. N. Adger, K. Brown, D. Conway, A. S. Halls, G. M. Pilling, J. D. Reynolds, N. L. Andrew, and N. K. Dulvy. 2009. Vulnerability of national economies to the impacts of climate change on fisheries. Fish and Fisheries 10(2):173-196. https://doi.org/10.1111/j.1467-2979.2008.00310.x

Allison, C., A. C. Winkler, A.-R. Childs, C. Muller, and W. M. Potts. 2023. Can social media platforms be used to foster improved environmental behaviour in recreational fisheries? Fisheries Research 258:106544. https://doi.org/10.1016/j.fishres.2022.106544

Anderson, M., and J. Jiang. 2018. Teens, social media and technology 2018. Pew Research Center, Washington, D.C., USA. https://www.pewinternet.org/2018/05/31/teens-social-media-technology-2018/

Angulo, E., and F. Courchamp. 2009. Rare species are valued big time. PLoS ONE 4(4):e5215. https://doi.org/10.1371/journal.pone.0005215

Arlinghaus, R., J. Alós, B. Beardmore, K. Daedlow, M. Dorow, M. Fujitani, D. Hühn, W. Haider, L. M. Hunt, B. M. Johnson, F. Johnston, T. Klefoth, S. Matsumura, C. Monk, T. Pagel, J. R. Post, T. Rapp, C. Riepe, H. Ward, and C. Wolter. 2017. Understanding and managing freshwater recreational fisheries as complex adaptive social-ecological systems. Reviews in Fisheries Science & Aquaculture 25(1):1-41. https://doi.org/10.1080/2330-8249.2016.1209160

Arlinghaus, R., S. J. Cooke, and W. Potts. 2013. Towards resilient recreational fisheries on a global scale through improved understanding of fish and fisher behaviour. Fisheries Management and Ecology 20(2-3):91-98. https://doi.org/10.1111/fme.12027

Azzurro, E., V. Sbragaglia, J. Cerri, M. Bariche, L. Bolognini, J. B. Souissi, G. Busoni, S. Coco, A. Chryssanthi, E. Fanelli, R. Ghanem, J. Garrabou, F. Gianni, F. Grati, J. Kolitari, G. Letterio, L. Lipej, C. Mazzoldi, N. Milone, F. Pannacciulli, A. Pešić, Y. Samuel-Rhoads, L. Saponari, J. Tomanic, N. E. Topçu, G. Vargiu, and P. Moschella. 2019. Climate change, biological invasions, and the shifting distribution of Mediterranean fishes: a large-scale survey based on local ecological knowledge. Global Change Biology 25(8):2779-2792. https://doi.org/10.1111/gcb.14670

Becken, S., B. Stantic, J. Chen, A. R. Alaei, and R. M. Connolly. 2017. Monitoring the environment and human sentiment on the Great Barrier Reef: assessing the potential of collective sensing. Journal of Environmental Management 203:87-97. https://doi.org/10.1016/j.jenvman.2017.07.007

Bellard, C., C. Bertelsmeier, P. Leadley, W. Thuiller, and F. Courchamp. 2012. Impacts of climate change on the future of biodiversity. Ecology Letters 15(4):365-377. https://doi.org/10.1111/j.1461-0248.2011.01736.x

Benoit, K., K. Watanabe, H. Wang, P. Nulty, A. Obeng, S. Müller, and A. Matsuo. 2018. quanteda: an R package for the quantitative analysis of textual data. Journal of Open Source Software 3:774. https://doi.org/10.21105/joss.00774

Berkes, F. 2011. Restoring unity: the concept of marine social-ecological systems. Pages 9-28 in R. E. Ommer, R. I. Perry, K. Cochrane, and P. Cury, editors. World fisheries: a social-ecological analysis. Wiley-Blackwell, Oxford, UK. https://doi.org/10.1002/9781444392241.ch2

Berkes, F., C. Folke, and J. Colding, editors. 1998. Linking social and ecological systems: management practices and social mechanisms for building resilience. Cambridge University Press, Cambridge, UK.

Bo, M., S. A. A. A. Mabruk, P. Balistreri, M. Bariche, I. E. Batjakas, F. Betti, M. Bilan, S. Canese, R. Cattaneo-Vietti, M. Corsini-Foka, F. Crocetta, A. Deidun, J. Dulčić, J. Grinyó, T. E. Kampouris, V. Ketsilis-Rinis, V. Kousteni, M. Koutsidi, H. Lubinevsky, S. Mavruk, C. Mytilineou, A. Petani, P. Puig, M. Salomidi, V. Sbragaglia, C. J. Smith, N. Stern, M. Toma, K. Tsiamis, B. Zava, and V. Gerovasileiou. 2020. New records of rare species in the Mediterranean Sea (October 2020). Mediterranean Marine Science 21(3):608-630.

Bonebrake, T. C., C. J. Brown, J. D. Bell, J. L. Blanchard, A. Chauvenet, C. Champion, I.-C. Chen, T. D. Clark, R. K. Colwell, F. Danielsen, A. I. Dell, J. M. Donelson, B. Evengård, S. Ferrier, S. Frusher, R. A. Garcia, R. B. Griffis, A. J. Hobday, M. A. Jarzyna, E. Lee, J. Lenoir, H. Linnetved, V. Y. Martin, P. C. McCormack, J. McDonald, E. McDonald-Madden, N. Mitchell, T. Mustonen, J. M. Pandolfi, N. Pettorelli, H. Possingham, P. Pulsifer, M. Reynolds, B. R. Scheffers, C. J. B. Sorte, J. M. Strugnell, M.-N. Tuanmu, S. Twiname, A. Vergés, C. Villanueva, E. Wapstra, T. Wernberg, and G. T. Pecl. 2018. Managing consequences of climate-driven species redistribution requires integration of ecology, conservation and social science. Biological Reviews 93(1):284-305. https://doi.org/10.1111/brv.12344

Burrows, M. T., D. Schoeman, L. B. Buckley, P. Moore, E. S. Poloczanska, K. M. Brander, C. Brown, J. F. Bruno, C. M. Duarte, B. S. Halpern, J. Holding, C. V. Kappel, W. Kiessling, M. I. O'Connor, J. M. Pandolfi, C. Parmesan, F. B. Schwing, W. J. Sydeman, and A. J. Richardson. 2011. The pace of shifting climate in marine and terrestrial ecosystems. Science 334(6056):652-655. https://doi.org/10.1126/science.1210288

Cheung, W. W. L., V. W. Y. Lam, J. L. Sarmiento, K. Kearney, R. Watson, and D. Pauly. 2009. Projecting global marine biodiversity impacts under climate change scenarios. Fish and Fisheries 10 (3):235-251. https://doi.org/10.1111/j.1467-2979.2008.00315.x

Cheung, W. W. L., R. Watson, and D. Pauly. 2013. Signature of ocean warming in global fisheries catch. Nature 497:365-368. https://doi.org/10.1038/nature12156

Correia, R. A., P. Jepson, A. C. M. Malhado, and R. J. Ladle. 2017. Internet scientific name frequency as an indicator of cultural salience of biodiversity. Ecological Indicators 78:549-555. https://doi.org/10.1016/j.ecolind.2017.03.052

Correia, R. A., R. Ladle, I. Jarić, A. C. M. Malhado, J. C. Mittermeier, U. Roll, A. Soriano-Redondo, D. Veríssimo, C. Fink, A. Hausmann, J. Guedes-Santos, R. Vardi, and E. D. Minin. 2021. Digital data sources and methods for conservation culturomics. Conservation Biology 35(2):398-411. https://doi.org/10.1111/cobi.13706

Di Minin, E., C. Fink, A. Hausmann, J. Kremer, and R. Kulkarni. 2021. How to address data privacy concerns when using social media data in conservation science. Conservation Biology 35 (2):437-446. https://doi.org/10.1111/cobi.13708

Đođo, Ž., N. Antolović, and J. Dulčić. 2016. New record of white grouper *Epinephelus aeneus* (Osteichthyes: Serranidae) in Croatian Adriatic waters. Pages 37-38 in Annales: Annals for Istrian and Meditterranean Studies. Series Historia Naturalis, Vol. 26, No. 1. Scientific and Research Center of the Republic of Slovenia, Piran, Slovenia. https://doi.org/10.19233/ASHN.2016.6

Dulčić, J., P. Tutman, and M. Ćaleta. 2006. Northernmost occurrence of the white grouper, *Epinephelus aeneus* (Perciformes: Serranidae), in the Mediterranean area. Acta Ichthyologica et Piscatoria 36(1):73-75. https://doi.org/10.3750/AIP2006.36.1.10

Fink, C., A. Hausmann, and E. Di Minin. 2020. Online sentiment towards iconic species. Biological Conservation 241:108289. https://doi.org/10.1016/j.biocon.2019.108289

Fogarty, H. E., M. T. Burrows, G. T. Pecl, L. M. Robinson, and E. S. Poloczanska. 2017. Are fish outside their usual ranges early indicators of climate-driven range shifts? Global Change Biology 23(5):2047-2057. https://doi.org/10.1111/gcb.13635

Francour, P., C. F. Boudouresque, J.-G. Harmelin, M. L. Harmelin-Vivien, and J.-P. Quignard. 1994. Are the Mediterranean waters becoming warmer? Information from biological indicators. Marine Pollution Bulletin 28:523-526. https://doi.org/10.1016/0025-326X(94)90071-X

García Molinos, J., B. S. Halpern, D. S. Schoeman, C. J. Brown, W. Kiessling, P. J. Moore, J. M. Pandolfi, E. S. Poloczanska, A. J. Richardson, and M. T. Burrows. 2016. Climate velocity and the future global redistribution of marine biodiversity. Nature Climate Change 6(1):83-88. https://doi.org/10.1038/nclimate2769

Glamuzina, B., P. Tutman, A. Geffen, V. Kozul, and B. Skaramuca. 2000. First record of white grouper, *Epinephelus aeneus* (Serranidae) in the South Eastern Adriatic. Cybium 24:306-308.

Gordoa, A., A. L. Dedeu, and J. Boada. 2019. Recreational fishing in Spain: first national estimates of fisher population size, fishing activity and fisher social profile. Fisheries Research 211:1-12. https://doi.org/10.1016/j.fishres.2018.10.026

Grolemund, G., and H. Wickham. 2011. Dates and times made easy with lubridate. Journal of Statistical Software 40:1-25. https://doi.org/10.18637/jss.v040.i03

Hunt, L. M., S. G. Sutton, and R. Arlinghaus. 2013. Illustrating the critical role of human dimensions research for understanding and managing recreational fisheries within a social-ecological system framework. Fisheries Management and Ecology 20 (2-3):111-124. https://doi.org/10.1111/j.1365-2400.2012.00870.x

Intergovernmental Panel on Climate Change (IPCC). 2021. Summary for policymakers. In V. Masson-Delmotte, P. Zhai, A. Pirani, S. L. Connors, C. Péan, S. Berger, N. Caud, Y. Chen, L. Goldfarb, M. I. Gomis, M. Huang, K. Leitzell, E. Lonnoy, J. B. R. Matthews, T. K. Maycock, T. Waterfield, O. Yelekçi, R. Yu, and B. Zhou, editors. Climate change 2021: the physical science basis. Contribution of Working Group I to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change. Cambridge University Press, Cambridge, UK. https://doi.org/10.1017/9781009157896.001

Jarić, I., C. A. Bellard, R. A. Correia, F. Courchamp, K. Douda, F. Essl, J. M. Jeschke, G. Kalinkat, L. Kalous, R. J. Lennox, A. Novoa, R. Proulx, P. Pyšek, A. Soriano-Redondo, A. T. Souza, R. Vardi, D. Veríssimo, and U. Roll. 2021. Invasion culturomics and iEcology. Conservation Biology 35(2):447-451. https://doi.org/10.1111/cobi.13707

Jarić, I., R. A. Correia, M. Bonaiuto, B. W. Brook, F. Courchamp, J. A. Firth, K. J. Gaston, T. Heger, J. M. Jeschke, R. J. Ladle, Y. Meinard, D. L. Roberts, K. Sherren, M. Soga, A. Soriano-Redondo, D. Veríssimo, and U. Roll. 2023. Transience of public attention in conservation science. Frontiers in Ecology and the Environment 21(7):333-340. https://doi.org/10.1002/fee.2598

Jarić, I., R. A. Correia, B. W. Brook, J. C. Buettel, F. Courchamp, E. Di Minin, J. A. Firth, K. J. Gaston, P. Jepson, G. Kalinkat, R. Ladle, A. Soriano-Redondo, A. T. Souza, and U. Roll. 2020b. iEcology: Harnessing large online resources to generate ecological insights. Trends in Ecology & Evolution 35(7):630-639. https://doi.org/10.1016/j.tree.2020.03.003

Jarić, I., U. Roll, R. Arlinghaus, J. Belmaker, Y. Chen, V. China, K. Douda, F. Essl, S. C. Jähnig, J. M. Jeschke, G. Kalinkat, L. Kalous, R. Ladle, R. J. Lennox, R. Rosa, V. Sbragaglia, K. Sherren, M. Šmejkal, A. Soriano-Redondo, A. T. Souza, C. Wolter, and R. A. Correia. 2020a. Expanding conservation culturomics and iEcology from terrestrial to aquatic realms. PLoS Biology 18(10):e3000935. https://doi.org/10.1371/journal.pbio.3000935

Jockers, M. 2020. Syuzhet: Extracts sentiment and sentiment-derived plot arcs from text.

Kapitza, K., H. Zimmermann, B. Martín-López, and H. von Wehrden. 2019. Research on the social perception of invasive species: a systematic literature review. NeoBiota 43:47-68. https://doi.org/10.3897/neobiota.43.31619

Katsanevakis, S., M. Coll, C. Piroddi, J. Steenbeek, F. Ben Rais Lasram, A. Zenetos, and A. C. Cardoso. 2014. Invading the Mediterranean Sea: biodiversity patterns shaped by human activities. Frontiers in Marine Science 1:32. https://doi.org/10.3389/fmars.2014.00032

Katsanevakis, S., A. Weber, C. Pipitone, M. Leopold, M. Cronin, M. Scheidat, T. Doyle, L. Buhl-Mortensen, P. Buhl-Mortensen, G. D'Anna, I. de Boois, P. Dalpadado, D. Damalas, F. Fiorentino, G. Garofalo, V. Giacalone, K. Hawley, Y. Issaris, J. Jansen, C. M. Knight, L. Knittweis, I. Kröncke, S. Mirto, I. Muxika, H. Reiss, H. R. Skjoldal, and S. Vöge. 2012. Monitoring marine populations and communities: review of methods and tools dealing with imperfect detectability. Aquatic Biology 16:31-52. https://doi.org/10.3354/ab00426

Kelly, R., K. Evans, K. Alexander, S. Bettiol, S. Corney, C. Cullen-Knox, C. Cvitanovic, K. de Salas, G. R. Emad, L. Fullbrook, C. Garcia, S. Ison, S. Ling, C. Macleod, A. Meyer, L. Murray, M. Murunga, K. L. Nash, K. Norris, M. Oellermann, J. Scott, J. S. Stark, G. Wood, and G. T. Pecl. 2022. Connecting to the oceans: supporting ocean literacy and public engagement. Reviews in Fish Biology and Fisheries 32(1):123-143. https://doi.org/10.1007/s11160-020-09625-9

Kueffer, C., and C. A. Kull. 2017. Non-native species and the aesthetics of nature. Pages 311-324 in M. Vilà and P. E. Hulme, editors. Impact of biological invasions on ecosystem services. Springer, Cham, Switzerland. https://doi.org/10.1007/978-3-319-45121-3_20

Ladle, R. J., R. A. Correia, Y. Do, G.-J. Joo, A. C. Malhado, R. Proulx, J.-M. Roberge, and P. Jepson. 2016. Conservation culturomics. Frontiers in Ecology and the Environment 14 (5):269-275. https://doi.org/10.1002/fee.1260

Lam, V. W. Y., W. W. L. Cheung, G. Reygondeau, and U. R. Sumaila. 2016. Projected change in global fisheries revenues under climate change. Scientific Reports 6:32607. https://doi.org/10.1038/srep32607

Lejeusne, C., P. Chevaldonné, C. Pergent-Martini, C. F. Boudouresque, and T. Pérez. 2010. Climate change effects on a miniature ocean: the highly diverse, highly impacted Mediterranean Sea. Trends in Ecology & Evolution 25(4):250-260. https://doi.org/10.1016/j.tree.2009.10.009

Lennox, R. J., V. Sbragaglia, K. W. Vollset, L. K. Sortland, L. McClenachan, I. Jarić, M. L. Guckian, K. Ferter, A. J. Danylchuk, S. J. Cooke, R. Arlinghaus, and W. M. Twardek. 2022. Digital fisheries data in the Internet age: emerging tools for research and monitoring using online data in recreational fisheries. Fish and Fisheries 23(4):926-940. https://doi.org/10.1111/faf.12663

Lionello, P., and L. Scarascia. 2018. The relation between climate change in the Mediterranean region and global warming. Regional Environmental Change 18(5):1481-1493. https://doi.org/10.1007/s10113-018-1290-1

Manfredo, M. J. 2008. Who cares about wildlife? Social science concepts for exploring human-wildlife relationships and conservation issues. Springer, New York, New York, USA. https://doi.org/10.1007/978-0-387-77040-6

Maxwell, S. M., E. L. Hazen, R. L. Lewison, D. C. Dunn, H. Bailey, S. J. Bograd, D. K. Briscoe, S. Fossette, A. J. Hobday, M. Bennett, S. Benson, M. R. Caldwell, D. P. Costa, H. Dewar, T. Eguchi, L. Hazen, S. Kohin, T. Sippel, and L. B. Crowder. 2015. Dynamic ocean management: defining and conceptualizing real-time management of the ocean. Marine Policy 58:42-50. https://doi.org/10.1016/j.marpol.2015.03.014

Melbourne-Thomas, J., A. Audzijonyte, M. J. Brasier, K. A. Cresswell, H. E. Fogarty, M. Haward, A. J. Hobday, H. L. Hunt, S. D. Ling, P. C. McCormack, T. Mustonen, K. Mustonen, J. A. Nye, M. Oellermann, R. Trebilco, I. van Putten, C. Villanueva, R. A. Watson, and G. T. Pecl. 2022. Poleward bound: adapting to climate-driven species redistribution. Reviews in Fish Biology and Fisheries 32(1):231-251. https://doi.org/10.1007/s11160-021-09641-3

Melbourne-Thomas, J., D. Tommasi, M. Gehlen, E. J. Murphy, J. Beckensteiner, F. Bravo, T. D. Eddy, M. Fischer, E. Fulton, M. Gogina, E. Hofmann, M. Ito, S. Mynott, K. Ortega-Cisneros, A. N. Osiecka, M. R. Payne, R. Saldívar-Lucio, and K. J. N. Scherrer. 2023. Integrating human dimensions in decadal-scale prediction for marine social-ecological systems: lighting the grey zone. ICES Journal of Marine Science 80(1):16-30. https://doi.org/10.1093/icesjms/fsac228

Monkman, G. G., M. Kaiser, and K. Hyder. 2018. The ethics of using social media in fisheries research. Reviews in Fisheries Science & Aquaculture 26(2):235-242. https://doi.org/10.1080/23308249.2017.1389854

Myers, N., R. A. Mittermeier, C. G. Mittermeier, G. A. B. da Fonseca, and J. Kent. 2000. Biodiversity hotspots for conservation priorities. Nature 403(6772):853-858. https://doi.org/10.1038/35002501

Nelder, J. A., and R. J. Baker. 2006. Generalized linear models. Encyclopedia of Statistical Sciences. https://doi.org/10.1002/0471667196.ess0866.pub2

Ojea, E., S. E. Lester, and D. Salgueiro-Otero. 2020. Adaptation of fishing communities to climate-driven shifts in target species. One Earth 2(6):544-556. https://doi.org/10.1016/j.oneear.2020.05.012

Ommer, R. E., R. Ian Perry, G. Murray, and B. Neis. 2012. Social-ecological dynamism, knowledge, and sustainable coastal marine fisheries. Current Opinion in Environmental Sustainability 4 (3):316-322. https://doi.org/10.1016/j.cosust.2012.05.010

Ostrom, E. 2009. A general framework for analyzing sustainability of social-ecological systems 325(5939):419-422. https://doi.org/10.1126/science.1172133

Parmesan, C., and G. Yohe. 2003. A globally coherent fingerprint of climate change impacts across natural systems. Nature 421:37-42. https://doi.org/10.1038/nature01286

Pecl, G. T., M. B. Araújo, J. D. Bell, J. Blanchard, T. C. Bonebrake, I.-C. Chen, T. D. Clark, R. K. Colwell, F. Danielsen, B. Evengård, L. Falconi, S. Ferrier, S. Frusher, R. A. Garcia, R. B. Griffis, A. J. Hobday, C. Janion-Scheepers, M. A. Jarzyna, S. Jennings, J. Lenoir, H. I. Linnetved, V. Y. Martin, P. C. McCormack, J.

McDonald, N. J. Mitchell, T. Mustonen, J. M. Pandolfi, N. Pettorelli, E. Popova, S. A. Robinson, B. R. Scheffers, J. D. Shaw, C. J. B. Sorte, J. M. Strugnell, J. M. Sunday, M.-N. Tuanmu, A. Vergés, C. Villanueva, T. Wernberg, E. Wapstra, and S. E. Williams. 2017. Biodiversity redistribution under climate change: impacts on ecosystems and human well-being. Science 355(6332). https://doi.org/10.1126/science.aai9214

Pecl, G. T., R. Kelly, C. Lucas, I. van Putten, R. Badhe, C. Champion, I.-C. Chen, O. Defeo, J. D. Gaitan-Espitia, B. Evengård, D. A. Fordham, F. Guo, R. Henriques, S. Henry, J. Lenoir, H. McGhie, T. Mustonen, S. Oliver, N. Pettorelli, M. L. Pinsky, W. Potts, J. Santana-Garcon, W. Sauer, A.-S. Stensgaard, M. W. Tingley, and A. Verges. 2023. Climate-driven 'species-on-the-move' provide tangible anchors to engage the public on climate change. People and Nature 5(5):1384-1402. https://doi.org/10.1002/pan3.10495

Pecl, G. T., E. Ogier, S. Jennings, I. van Putten, C. Crawford, H. Fogarty, S. Frusher, A. J. Hobday, J. Keane, E. Lee, C. MacLeod, C. Mundy, J. Stuart-Smith, and S. Tracey. 2019. Autonomous adaptation to climate-driven change in marine biodiversity in a global marine hotspot. Ambio 48(12):1498-1515. https://doi.org/10.1007/s13280-019-01186-x

Perry, A. L., P. J. Low, J. R. Ellis, and J. D. Reynolds. 2005. Climate change and distribution shifts in marine fishes. Science 308 (5730):1912-1915. https://doi.org/10.1126/science.1111322

Perry, R. I., R. E. Ommer, M. Barange, and F. Werner. 2010. The challenge of adapting marine social-ecological systems to the additional stress of climate change. Current Opinion in Environmental Sustainability 2(5-6):356-363. https://doi.org/10.1016/j.cosust.2010.10.004

Peters, G. P., R. M. Andrew, T. Boden, J. G. Canadell, P. Ciais, C. Le Quéré, G. Marland, M. R. Raupach, and C. Wilson. 2013. The challenge to keep global warming below 2 °C. Nature Climate Change 3(1):4-6. https://doi.org/10.1038/nclimate1783

Philippart, C. J. M., R. Anadón, R. Danovaro, J. W. Dippner, K. F. Drinkwater, S. J. Hawkins, T. Oguz, G. O'Sullivan, and P. C. Reid. 2011. Impacts of climate change on European marine ecosystems: observations, expectations and indicators. Journal of Experimental Marine Biology and Ecology 400(1-2):52-69. https://doi.org/10.1016/j.jembe.2011.02.023

Pinsky, M. L., and M. Fogarty. 2012. Lagged social-ecological responses to climate and range shifts in fisheries. Climatic Change 115(3):883-891. https://doi.org/10.1007/s10584-012-0599-x

Pinsky, M. L., R. L. Selden, and Z. J. Kitchel. 2020. Climate-driven shifts in marine species ranges: scaling from organisms to communities. Annual Review of Marine Science 12(1):153-179. https://doi.org/10.1146/annurev-marine-010419-010916

Pita, P., K. Hyder, P. Gomes, C. Pita, M. Rangel, P. Veiga, J. Vingada, and S. Villasante. 2018. Economic, social and ecological attributes of marine recreational fisheries in Galicia, Spain. Fisheries Research 208:58-69. https://doi.org/10.1016/j.fishres.2018.07.014

Plutchik, R. 1980. A general psychoevolutionary theory of emotion. Pages 3-33 in R. Plutchik and H. Kellerman, editors. Theories of emotion. Academic Press, New York, New York, USA. https://doi.org/10.1016/B978-0-12-558701-3.50007-7

Pollard, D., P. Francour, and S. Fennessy. 2018. *Epinephelus aeneus*. The IUCN Red List of Threatened Species 2018:e. T132722A100459597. http://dx.doi.org/10.2305/IUCN.UK.2018-2.RLTS.T132722A100459597.en

Poloczanska, E. S., C. J. Brown, W. J. Sydeman, W. Kiessling, D. S. Schoeman, P. J. Moore, K. Brander, J. F. Bruno, L. B. Buckley, M. T. Burrows, C. M. Duarte, B. S. Halpern, J. Holding, C. V. Kappel, M. I. O'Connor, J. M. Pandolfi, C. Parmesan, F. Schwing, S. A. Thompson, and A. J. Richardson. 2013. Global imprint of climate change on marine life. Nature Climate Change 3 (10):919-925. https://doi.org/10.1038/nclimate1958

Reinecke, L., and S. Trepte. 2014. Authenticity and well-being on social network sites: a two-wave longitudinal study on the effects of online authenticity and the positivity bias in SNS communication. Computers in Human Behavior 30:95-102. https://doi.org/10.1016/j.chb.2013.07.030

Riutort, J. 2012. Premier signalement d'*Epinephelus aeneus* (Geoffroy St.Hilaire, 1817) (Perciformes, Serranidae) dans les eaux méditerranéennes françaises. Bulletin de la Société des Sciences Historiques et Naturelles de la Corse 738-739:183-192.

Salat, J., J. Pascual, M. Flexas, T. M. Chin, and J. Vazquez-Cuervo. 2019. Forty-five years of oceanographic and meteorological observations at a coastal station in the NW Mediterranean: a ground truth for satellite observations. Ocean Dynamics 69 (9):1067-1084. https://doi.org/10.1007/s10236-019-01285-z

Sbragaglia, V., J. W. Brownscombe, S. J. Cooke, A. D. Buijse, R. Arlinghaus, and W. M. Potts. 2023a. Preparing recreational fisheries for the uncertain future: an update of progress towards answering the 100 most pressing research questions. Fisheries Research 263:106662. https://doi.org/10.1016/j.fishres.2023.106662

Sbragaglia, V., S. Coco, R. A. Correia, M. Coll, and R. Arlinghaus. 2021a. Analyzing publicly available videos about recreational fishing reveals key ecological and social insights: a case study about groupers in the Mediterranean Sea. Science of The Total Environment 765:142672. https://doi.org/10.1016/j.scitotenv.2020.142672

Sbragaglia, V., R. A. Correia, S. Coco, and R. Arlinghaus. 2020. Data mining on YouTube reveals fisher group-specific harvesting patterns and social engagement in recreational anglers and spearfishers. ICES Journal of Marine Science 77(6):2234-2244. https://doi.org/10.1093/icesjms/fsz100

Sbragaglia, V., R. A. Correia, and E. Di Minin. 2021b. Responsible use of social media data is needed: a reply to Maya-Jariego et al. "Plenty of black money: Netnography of illegal recreational underwater fishing in southern Spain." Marine Policy 134:104780. https://doi.org/10.1016/j.marpol.2021.104780

Sbragaglia, V., L. Espasandín, S. Coco, A. Felici, R. A. Correia, M. Coll, and R. Arlinghaus. 2022. Recreational angling and spearfishing on social media: insights on harvesting patterns, social engagement and sentiments related to the distributional range shift of a marine invasive species. Reviews in Fish Biology and Fisheries 32(2):687-700. https://doi.org/10.1007/s11160-022-09699-7

Sbragaglia, V., L. Espasandín, I. Jarić, R. Vardi, F. Ramírez, and M. Coll. 2023b. Tracking ongoing transboundary marine

distributional range shifts in the digital era. Marine Ecology Progress Series. https://doi.org/10.3354/meps14309

Shackleton, R. T., B. M. H. Larson, A. Novoa, D. M. Richardson, and C. A. Kull. 2019a. The human and social dimensions of invasion science and management. Journal of Environmental Management 229:1-9. https://doi.org/10.1016/j.jenvman.2018.08.041

Shackleton, R. T., D. M. Richardson, C. M. Shackleton, B. Bennett, S. L. Crowley, K. Dehnen-Schmutz, R. A. Estévez, A. Fischer, C. Kueffer, C. A. Kull, E. Marchante, A. Novoa, L. J. Potgieter, J. Vaas, A. S. Vaz, and B. M. H. Larson. 2019b. Explaining people's perceptions of invasive alien species: a conceptual framework. Journal of Environmental Management 229:10-26. https://doi.org/10.1016/j.jenvman.2018.04.045

Spijkers, J., and W. J. Boonstra. 2017. Environmental change and social conflict: the northeast Atlantic mackerel dispute. Regional Environmental Change 17(6):1835-1851. https://doi.org/10.1007/s10113-017-1150-4

Toivonen, T., V. Heikinheimo, C. Fink, A. Hausmann, T. Hiippala, O. Järv, H. Tenkanen, and E. Di Minin. 2019. Social media data for conservation science: a methodological overview. Biological Conservation 233:298-315. https://doi.org/10.1016/j.biocon.2019.01.023

Townhill, B. L., Z. Radford, G. Pecl, I. van Putten, J. K. Pinnegar, and K. Hyder. 2019. Marine recreational fishing and the implications of climate change. Fish and Fisheries 20(5):977-992. https://doi.org/10.1111/faf.12392

Vitale, G., A. L. Dedeu, M. Pujol, and V. Sbragaglia. 2021. Characterizing the profile of recreational fishers who share their catches on social media. Frontiers in Marine Science 8:1663. https://doi.org/10.3389/fmars.2021.768047

Wang, S., Z. Leviston, M. Hurlstone, C. Lawrence, and I. Walker. 2018. Emotions predict policy support: Why it matters how people feel about climate change. Global Environmental Change 50:25-40. https://doi.org/10.1016/j.gloenvcha.2018.03.002

Ward, H. G. M., M. S. Allen, E. V. Camp, N. Cole, L. M. Hunt, B. Matthias, J. R. Post, K. Wilson, and R. Arlinghaus. 2016. Understanding and managing social-ecological feedbacks in spatially structured recreational fisheries: the overlooked behavioral dimension. Fisheries 41(9):524-535. https://doi.org/10.1080/03632415.2016.1207632

Zimmer, M. 2010. "But the data is already public": on the ethics of research in Facebook. Ethics and Information Technology 12 (4):313-325. https://doi.org/10.1007/s10676-010-9227-5